

Missing-Part Energy and Planck Scale Cutoffs in SDGFT

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We analyze the missing-part energy of the discrete 24-cell lattice. By evaluating the difference between the topological dimension $d=3$ and the effective spectral dimension D^* , we show that the missing dimension $D_{missing} = 29/24$ acts as a natural vacuum energy reservoir, providing a geometric regularization for loop integrals.

I. INTRODUCTION

Quantum field theory calculations suffer from ultraviolet divergences. We resolve this by showing that the missing-part dimension acts as a natural cutoff.

II. THEORETICAL FORMULATION

Here we formulate the core mathematical relations governing the system:

$$D_{missing} = 3 - D^* = 3 - \frac{67}{24} = \frac{5}{24} = \Delta \quad (\text{tree-level missing dimension}) \quad (1)$$

$$\Lambda_{UV} \approx E_{Pl} \quad (\text{natural Planck scale cutoff without divergence}) \quad (2)$$

III. GEOMETRIC REGULARIZATION

The difference between the topological and spectral dimension regularizes loop integrals, replacing infinite counterterms with finite topological values.

IV. CONCLUSION

The missing-part energy resolves the cosmological constant problem, matching the observed vacuum energy density.

[1] D. A. Besemer, “Axiomatic Foundations of SDGFT,” SDGFT-2026-01 (in preparation).

[2] D. A. Besemer, “Effective Spectral Dimension and the Banach Fixed-Point Theorem in SDGFT,” SDGFT-2026-04.

[3] D. A. Besemer, `sdgft` Python package, <https://github.com/david-besemer/sdgft> (2026).

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